

Crypto Price Tracker Dashboard – Full-Stack Hackathon Challenge

📌 Objective

Build a full-stack web application that displays **live prices of BTC, ETH, and BNB**, refreshed every **10 seconds**, showing:

- Prices in **INR** (converted from USD using a free exchange rate API)
- **Price direction** (up/down from last fetch)
- **% change**
- Clean UI updates

The backend handles all API communication; the frontend displays the data dynamically.

📌 Tech Stack

Layer	Tech
Frontend	React (hooks, functional components)
Backend	Node.js + Express
APIs	CoinGecko (crypto prices), ExchangeRate.host (USD → INR)
Charts	Not required, but simple cards or UI indicators

📁 Folder Structure

```
bash
```

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/crypto-tracker

/client → React frontend

/server → Node.js backend

README.md → this file

📁 Backend Requirements (/server)

1. GET /api/prices

Returns current prices for BTC, ETH, and BNB with values in **INR**.

☒ What it should do:

- Fetch **USD prices** from CoinGecko:
 - bash
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 - https://api.coingecko.com/api/v3/simple/price?ids=bitcoin,ethereum,binancecoin&vs_currencies=usd
- Fetch **USD to INR** exchange rate from:
 - arduino
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 - <https://api.exchangerate.host/latest?base=USD&symbols=INR>
- Convert crypto prices to INR
- Return this JSON:

json

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```
{
  "BTC": {
    "usd": 63123.45,
    "inr": 5239226.35
  },
  "ETH": {
    "usd": 3523.11,
    "inr": 292418.13
  },
  "BNB": {
```

```
    "usd": 600.50,  
    "inr": 49841.50  
  },  
  "lastUpdated": "2025-07-11T15:23:50.000Z"  
}
```

⌚ Optional Optimization:

- Use an in-memory cache for 10 seconds to avoid hitting APIs more than once per interval (optional but good to show logic understanding)

📱 Frontend Requirements (/client)

Dashboard View:

For each coin (BTC, ETH, BNB), show:

- Coin Name
- Price in INR (e.g., ₹5,23,922.35)
- Arrow icon:
 - ↑ if price went up
 - ↓ if price went down
- % change since last check
- Timestamp: "Last updated: 3:23:50 PM"

Example:

```
sql  
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BTC  
₹5,23,922.35 ↑ +1.52%  
Last updated: 3:23:50 PM
```

🔄 Refresh Logic:

- Fetch `/api/prices` every **10 seconds**
- Compare new prices with previous prices in component state
- Show price direction and % change accurately

- `js`
- `CopyEdit`
- `const percentChange = ((newPrice - oldPrice) / oldPrice) * 100;`

☒ UI Notes:

- Use any component library or just CSS
- Must work well on desktop
- Optional: Color-code up/down (green/red)

☒ Constraints

- ✗ Do not fetch data directly from frontend (must use `backend /api/prices`)
- ✗ Do not use database or `localStorage`
- ☒ Use only free public APIs mentioned
- ☒ Keep INR format (use `Intl.NumberFormat` or similar)

Time Limit

3 hours

☒ Evaluation Criteria

Area	Evaluation Focus
Frontend	React hooks, polling logic, UI state changes, rendering
Backend	API integration, response formatting, error handling
Logic	Correct INR conversion, % change calculation
Code Quality	Folder structure, naming, readability
Bonus	INR/USD switch, color indicators, error UI

(Optional)